

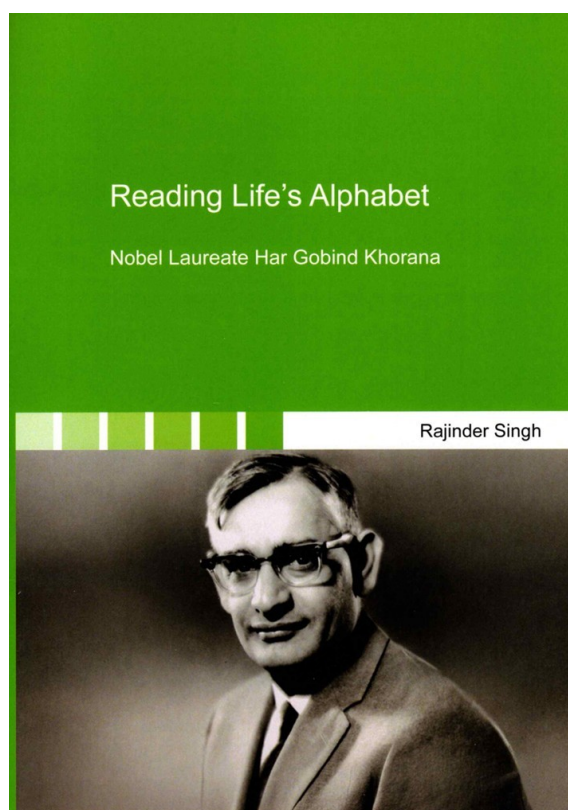
Book Review: **Reading Life's Alphabet: Nobel Laureate Har Gobind Khorana**

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Rajinder Singh's journey in history of science starts in 1995 at University of Oldenburg in Germany. He is an authority on Indian history of science with 57 books to his credit. The book under review describes the life and achievements of Nobel Laureate, Har Gobind Khorana. In the Preface, author reveals that a book on Har Gobind Khorana has been published for School and College students under the title, "*Cracking Life's Code: The Journey of Nobel Laureate Har Gobind Khorana*", by Punjab State Council for Science & Technology, Chandigarh (2026).

Chandrima Shaha writes in her Foreword: "Among the scientists who have shaped our world, Har Gobind Khorana stands out for his pioneering work in molecular biology and biotechnology. As an Indian-American molecular biologist, Khorana deciphered the genetic code, fundamentally advancing science. He received the 1968 Nobel Prize in Physiology or

Medicine for revealing how RNA's nucleotide sequence directs protein synthesis. This book explores Khorana's life and the lasting impact of his contributions".

Prof. Arun Kumar Grover in his Prologue has traced the history of Har Gobind Khorana's contemporaries under the heading, "*Har Gobind Khorana and his Contemporaries at Lahore*". It is most fascinating introduction to Khorana's era of Punjab University, Lahore; his brilliant contemporaries and their frustrations after return to India to find suitable jobs after Partition of India in 1947. Grover vividly describes the frustration faced by Har Gobind Khorana on his failure to get a job commensurate with his training abroad. Ultimately, he decided to return to Cambridge to join Prof. Alexander Todd's Laboratory after getting a Nuffield Foundation Fellowship.

In the Introduction, the author writes about special features of his book, claiming that this version of Khorana's biography shows unknown traits of his life; and fills the gap left in previous biographies. In Chapter 1, Rajinder Singh (RS) writes about Khorana's family background, his education from village Primary School to Govt. College, Lahore, and his passing out B.Sc. (Hons.) and M.Sc. in Chemistry in 1943 and 1945, respectively from Punjab University, Lahore. He was a scholarship holder during his university education.

In Chapter 2, RS describes the research work done by Khorana as part of his Ph.D. thesis in Liverpool and during his stay in Cambridge. He wrote his Ph.D. thesis on "*Chemistry of violacein and related topics – Isolation of alkaloids from Daphnandra species*". Khorana observed that the chemistry of violacein is still not well-understood. He described an improved method for cultivating *Chromobacterium violaceum* and isolating violacein in a crystalline form. After a brief stay in India, Khorana returned to Cambridge to work on Peptides. In a review article "*Structural Investigation of Peptides and Proteins*," Khorana outlined the progress in peptide and protein structure analysis, attributing advancements to improved analytical techniques. He emphasized the importance of isolating pure peptides, identifying amino acids, and determining their sequence.

Chapter 3, "*The Emergence of DNA and Cell Science*", is an important milestone in the history of development of Molecular Biology and Genetic Engineering. RS traces the history of DNA and RNA and determination of their helical structure: "Maurice Wilkins and Rosalind Franklin, working at King's College London in the early 1950s, played crucial roles in uncovering the structure of DNA. Franklin used X-ray crystallography to produce highly detailed images of DNA fibres. One of her key results, the famous "Photo 51", revealed the

helical nature of DNA”. Before Khorana embarked on his Noble Prize-winning research, the field was hot and ready for new discoveries.

Chapter 4 describes the research work done by Khorana during his decade long stay in UBC. He studied Carbodiimides and their synthesis as reported in his Review paper: “In 1953, Khorana published a comprehensive review entitled “*Chemistry of Carbodiimides*,” drawing on earlier German literature encountered during his work in Zurich. Khorana’s review covered methods of synthesis (e.g., from thioureas), physical and chemical properties, and comparisons with related compounds. He discussed their practical uses in nucleotide and sulfonic anhydride synthesis, peptide end-group identification”. Khorana was the first to demonstrate that carbodiimides can be used in the synthesis of nucleotides.

Khorana made important contributions to the study of energy-rich phosphate esters. He developed synthetic methods for producing poly-nucleosides and later extended this work to cyclic phosphate esters. In the period from 1956 to 1958, Khorana’s laboratory published six papers on “Phosphorylated sugars” in Journal of the American Chemical Society. RS narrates two success stories of Har Gobind Khorana towards the end of this Chapter. In August 1954, various newspapers recorded about Khorana’s success for making a new chemical that would significantly accelerate the treatment of cancer. His second success story related to *Study of Nucleoside and the Making of Coenzyme A*. The Victoria Daily reported: “Pride of the council is the work being done by Dr. H. G. Khorana, whose research into nucleic acids- the basic stuff of living organism – has excited attention around the world”. Despite his reputation touching the skies in UBC, Khorana opted to join Wisconsin at a higher salary and in view of better facilities.

Chapter 5 describes Khorana’s research at Enzyme Institute in US. RS writes: “In 1960, Khorana accepted a position as co-director of the Enzyme Research Institute, University of Wisconsin, Madison. Two years later, he became a professor of biochemistry. In 1964, he was Conrad A. Elvehjem Professor of Life Sciences”. The two most important investigations done by Khorana group in Wisconsin relate to “*Breaking the Codon Code or Revealing How Codons Specify Amino Acids*”, and, “*Making of the First Biologically Active Synthetic Gene*”.

It is very fascinating to read the story of path-breaking research carried out by Har Gobind Khorana in Wisconsin: “In 1968, Khorana wrote a long article on “Nucleic Acid Synthesis.” He detailed his and his colleagues’ work on nucleic acid synthesis, emphasizing the

importance of chemically defined polynucleotides for studying genetic information. Khorana reviewed key milestones - after the discovery of DNA's structure, DNA polymerase, and RNA polymerase, scientists realized synthetic polynucleotides could direct protein synthesis in cell-free systems, inspiring chemists to make defined sequences to explore gene control. Khorana and his team's landmark paper in genetics was published in 1970 in *Nature*".

Before he left Wisconsin to join MIT Harvard, Khorana was highly acclaimed for his research work in local media. To sum up, H. G. Khorana made key contributions to understanding the genetic code and how DNA directs protein synthesis. He showed that the code is made of triplet sequences, is degenerate, and is universal across organisms by synthesizing RNA with repeating sequences. He also advanced knowledge of DNA polymerases, clarifying their roles in replication, mutations, DNA repair, and disease processes.

Chapter 6 sums up the process of awarding Nobel Prize to Har Gobind Khorana. It is an interesting story recorded by Rajinder Singh based on his findings in the archival cell of the Nobel Foundation. Khorana was nominated for the Nobel Prize (NP) in Chemistry in 1963, 1965, 1966, 1967 and 1968 by various nominators from India and Europe. In 1968, Khorana received maximum 6 nominations, including 3 from Physiology or Medicine Nobel Laureates of France, awarded in 1965. Nobel Committee (NC) decided to award NP to Har Gobind Khorana jointly along with Marshall W. Nirenberg and R.W. Holley.

NC expert P. Reichard noted that the three scientists had made fundamental contributions to defining the chemical basis of heredity, achievements of major significance for biology, chemistry, and medicine. The NP committee reaffirmed that the deciphering of the genetic code represented a discovery of exceptional importance and should be recognized with a Nobel Prize. It also noted that the Nobel Committee for Physiology or Medicine was simultaneously considering awarding these same researchers for their contributions. There is no doubt that the Nobel Committee for Physiology or Medicine saw the work of the three chemists as worth of the Nobel Prize and recommended their names. The Royal Swedish Academy of Science, which takes the final decision regarding the Nobel Prize, decided to divide the prize in three parts. Each nominee got 1/3rd of the Nobel Prize "*for their interpretation of the genetic code and its function in protein synthesis.*"

Chapter 7 describes the work of H.G. Khorana carried out at MIT: "In 1970, Khorana was offered the Alfred P. Sloan Professor of Biology and Chemistry at the Massachusetts Institute of Technology. He changed his focus to the study of membranes and signal transduction.

Until his retirement in 2007, he worked on these topics”. Some of his important investigations relate to *Synthesis of the First Biologically Active tRNA Gene and Cloning; Study of Bacteriorhodopsin; The Bacteriorhodopsin Gene; Codon Replacement in the Bacterio-opsin Gene; Study of G Protein; Purification of Rhodopsin from Mammalian Cells; and Stabilizing Rhodopsin with Designer Peptides*. To sum up, Khorana’s work on G protein made it possible to understand how GPCRs activate G proteins at the molecular level. It shaped modern signal transduction research.

Chapter 8 is simply based on peer opinions and professional interactions of Har Gobind Khorana. The author opines that Khorana could win the NP due to his immigration from India to foreign countries where he got good facilities for research and opportunities to interact with Nobel Laureates such as F. Crick, J. D. Watson, J. Lederberg, and A. Kornberg. This correspondence is long but informative about exchange of *ideas* between Khorana and the Nobel Laureates. A sample correspondence with F. Crick is as follows: “Khorana told Frick that the final goal is to create a gene capable of in vitro transcription using its own signals, which would allow systematic modifications of the structural gene and testing synthetic promoters and terminators on other DNA sequences. Additionally, Khorana described his growing interest in membrane chemistry, including collaborations with Efraim Racker’s group on reconstituting membrane functions”.

Chapter 9 is focussed on “Honours and Awards” received by Khorana. H. G. In honour of his research carried out by Har Gobind Khorana in UBC, Khorana Chair in Biological Chemistry was created at the UBC after he won the NP. Donations were collected from old Scholars of Khorana, BC Government, and industry to create a minimum endowment of one million dollars for Khorana Chair. Michel Smith, a research collaborator of Khorana, was mainly responsible for Fund raising for the Chair. The author concludes: “Smith concluded by observing that while the ability to give is a form of good fortune, but the deeper “good luck” he cherishes is having met and worked with Gobind Khorana beginning in 1956 and having UBC as the home of his professional career—experiences for which he remains profoundly grateful”. On page 139, RS has given an elaborate list of Awards, Medals and Prizes won by Har Gobind Khorana.

Chapter 10 “Concluding Remarks” sums up the philosophy of science research of Har Gobind Khorana: “There is a lot of serendipity in science; the more significant things in one’s research happen when something has been “naively” overlooked. For example, an organic

chemist puts to test his mastery of organic chemistry by designing multistep syntheses. Yet, the exciting moments are those when the “unexpected” happens. ... different phases of my own work fully support this observation.” Khorana was a great mentor in research. He left everlasting impression in the world. He trained over 150 postdoctoral fellows and several graduate students, who influenced the development of gene technology.

I have reviewed nearly a dozen books authored by Rajinder Singh but this one proved to be a hard nut to crack. In my student life, Organic Chemistry was my most hated subject. Har Gobind Khorana’s research covered a broad spectrum of Organic synthesis which I failed to comprehend. I wonder how Rajinder Singh collected so much archival material on Khorana to make his biography a rare feature in history of science. I recommend this biography for Indian students, in general, and Punjabi students, in particular, to learn about the vicissitudes of getting a Nobel Prize by our Indo-American scientist.